

METAL SOLUTIONS

EOS StainlessSteel PH1

Material Data Sheet

EOS STAINLESSSTEEL PH1

EOS StainlessSteel PH1 is a pre-alloyed stainless steel in fine powder form. The chemistry of EOS StainlessSteel PH1 conforms to the compositions of DIN 1.4540 and UNS S15500. This kind of steel is characterized by having good corrosion resistance and excellent mechanical properties, especially in the precipitation hardened state. This type of steel is widely used in variety of engineering applications requiring high hardness, strength and corrosion resistance. This material is ideal for many part-building applications (DirectPart) such as functional metal prototypes, small series products, individualised products or spare parts. Standard processing parameters use full melting of the entire geometry with 20 µm layer thickness. Using standard parameters the mechanical properties are fairly uniform in all directions. Parts made from EOS StainlessSteel PH1 can be machined, spark-eroded, welded, micro shot-peened, polished and coated if required.

MAIN CHARACTERISTICS

- High strength and hardness
- Corrosion resistant
- For mildly corrosive environments

TYPICAL APPLICATIONS

- Engineering applications including functional prototypes, small series products, individualised products or spare parts
- Parts requiring high corrosion resistance, sterilisability, etc.
- Parts requiring particularly high hardness and strength

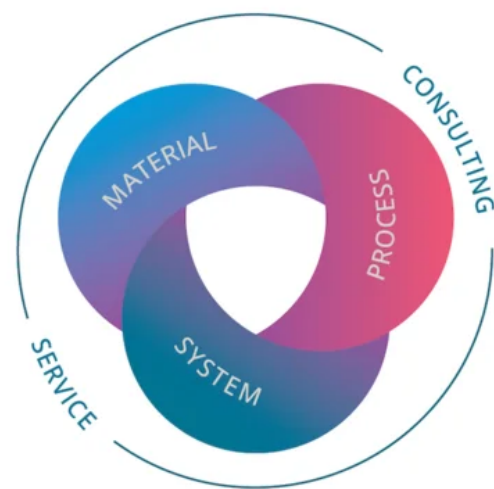
The EOS Quality Triangle

EOS uses an approach that is unique in the AM industry, taking each of the three central technical elements of the production process into account: the system, the material and the process. The data resulting from each combination is assigned a Technology Readiness Level (TRL) which makes the expected performance and production capability of the solution transparent.

EOS incorporates these TRLs into the following two categories:

- Premium products (TRL 7-9): offer highly validated data, proven capability and reproducible part properties.
- Core products (TRL 3 and 5): enable early customer access to newest technology still under development and are therefore less mature with less data.

All of the data stated in this material data sheet is produced according to EOS Quality Management System and international standards



POWDER PROPERTIES

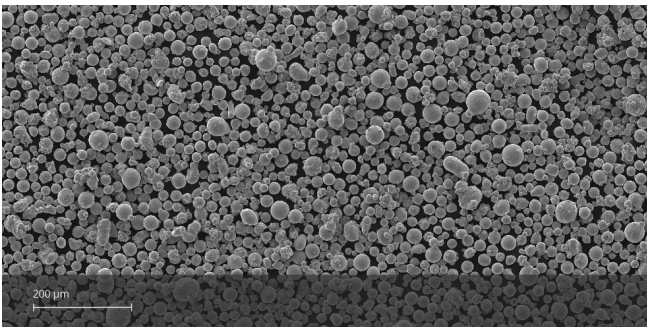
The chemistry of EOS StainlessSteel PH1 conforms to the compositions of DIN 1.4540 and UNS S15500.

Powder Chemical Composition (wt.-%)

Element	Min.	Max.
Cr	14	15.5
Ni	3.5	5.5
Cu	2.5	4.5
Mn	0	1
Si	0	1
C	0	0.07
Mo	0	0.5
Nb	0.15	0.45

Powder Particle Size

Generic Particle Size Distribution	15 - 55 µm
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HEAT TREATMENT

Description

EOS PH1 can be heat treated to match various needs of different applications. The two step heat treatment can be performed under vacuum or inert gas atmosphere. First step is solution annealing to minimize amount of austenite in the martensitic matrix. The needed hardness and strength is achieved through aging treatment where hardening takes place through forming of precipitates.

Steps

Solution annealing:
0.5-1 h at 1040 °C (±15 °C) measured from the part followed by rapid air cooling to room temperature (below 32 °C). Cooling rate 5-60 °C/min. Reaching room temperature before starting aging treatment is required to achieve desired microstructure.

Ageing:
Peak hardness of 45 HRC is achieved with H900 ageing treatment of 1 h at 480 °C (±10 °C) measured from the part followed by air cooling. Mechanical properties presented in this document achieved through this aging procedure. For lower hardness and strength choose aging procedure according to the table.

Condition	Ageing temperature [°C]	Hold time [h]	Cooling	Hardness [HRC]
H900	480	1	Air	45
H925	495	4	Air	42
H1025	550	4	Air	37
H1075	580	4	Air	35
H1100	595	4	Air	33
H1150	620	4	Air	32

EOS StainlessSteel PH1 for EOS M 290 | 40/80 µm

EOS M 290 - 40 µm - TRL 3

Process Information Metal

This process product is optimized for flexible and fast production of PH1 parts with EOS M 290 and M 400-4 systems. The parameter set has three different layer thickness options that can all be utilized within the same build: 40µm, 80 µm and 40/80 µm Skin. For high productivity needs a 80 µm parameter set is included with a build rate more than double the long established 40 µm parameter set. Both can be used separately for different parts or combined by using 40/80 µm Skin with faster 80 µm for the core of the part and higher quality 40 µm building for the surface of the part with defined thickness. Sectionioning parts in vertical direction for different parameters is also possible.

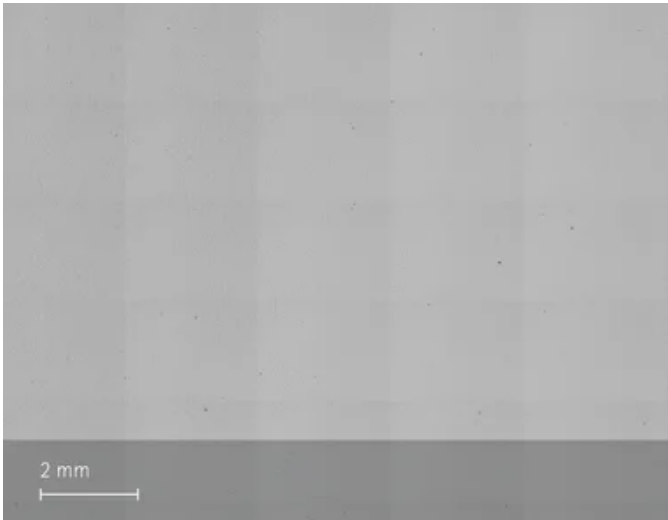
System Setup	EOS M 290
EOS Material set	PH1_040_080_Core M291
Software Requirements	EOSPRINT 2.11 or newer EOSYSTEM 2.14 or newer
Recoater Blade	Ceramic
Nozzle	EOS Grid Nozzle
Inert gas	Nitrogen
Sieve	63 µm

Additional Information	
Layer Thickness	40 µm, 80 µm, 40/80 µm
Volume Rate	4.1-7.9 mm³/s
Typical Dimensional Change after HT [%]	+0.1 %

Chemical and Physical Properties of Parts



Micrograph of polished surface. As manufactured 40µm



Micrograph of polished surface. As manufactured 80µm

Microstructure of the Produced Parts

Defects	Thickness	Result	Number of Samples
Average Defect Percentage	40 µm	<0.1 %	-
Average Defect Percentage	80 µm	<0.2 %	-

Mechanical Properties

Mechanical Properties Heat Treated

EN ISO 6892-1 Room Temperature 40 µm	Yield Strength [MPa]	Tensile Strength [MPa]	Elongation at Break A [%]	Reduction of Area Z [%]	Modulus of elasticity [GPa]	Number of Samples
Vertical	1305	1440	8	-	-	-
Horizontal	1300	1450	10	-	-	-

H900 heat treatment

Mechanical Properties Heat Treated

EN ISO 6892-1 Room Temperature 80 µm	Yield Strength [MPa]	Tensile Strength [MPa]	Elongation at Break A [%]	Reduction of Area Z [%]	Modulus of elasticity [GPa]	Number of Samples
Vertical	1310	1440	7	-	-	-
Horizontal	1310	1460	10	-	-	-

H900 heat treatment

Impact Toughness

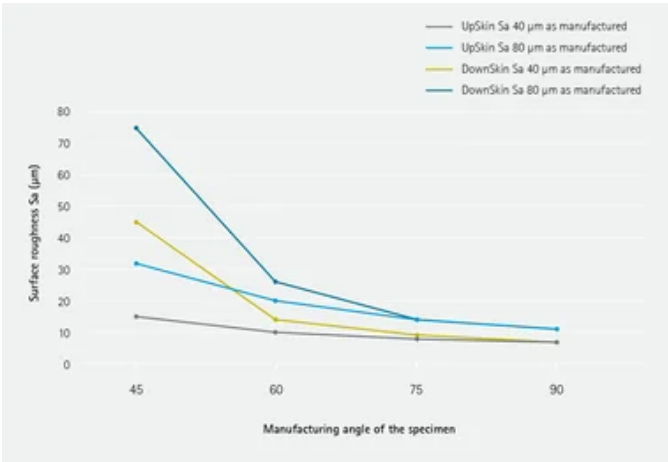
EN ISO 148-1, Charpy-V Heat Treated	
Typical Impact Toughness [J]	8
Layer Thickness	40 µm

H900 heat treatment

EN ISO 148-1, Charpy-V Heat Treated	
Typical Impact Toughness [J]	6
Layer Thickness	80 µm

H900 heat treatment

Surface Roughness



Coefficient of Thermal Expansion

ASTM E228	Temperature
10.6*10 ⁻⁶ /K	25 – 100 °C
11.1*10 ⁻⁶ /K	25 – 200 °C
11.4*10 ⁻⁶ /K	25 – 300 °C
11.7*10 ⁻⁶ /K	25 – 400 °C

EOS StainlessSteel PH1 for EOS M 400-4 | 40/80 µm

EOS M 400-4 - 40 µm - TRL 3

Process Information Metal

This process product is optimized for flexible and fast production of PH1 parts with EOS M 400-4 system. The parameter set has three different layer thickness options that can all be utilized within the same build: 40µm, 80 µm and 40/80 µm Skin. For high productivity needs a 80 µm parameter set is included with a build rate more than double the long established 40 µm parameter set. Both can be used separately for different parts or combined by using 40/80 µm Skin with faster 80 µm for the core of the part and higher quality 40 µm building for the surface of the part with defined thickness. Sectionioning parts in vertical direction for different parameters is also possible.

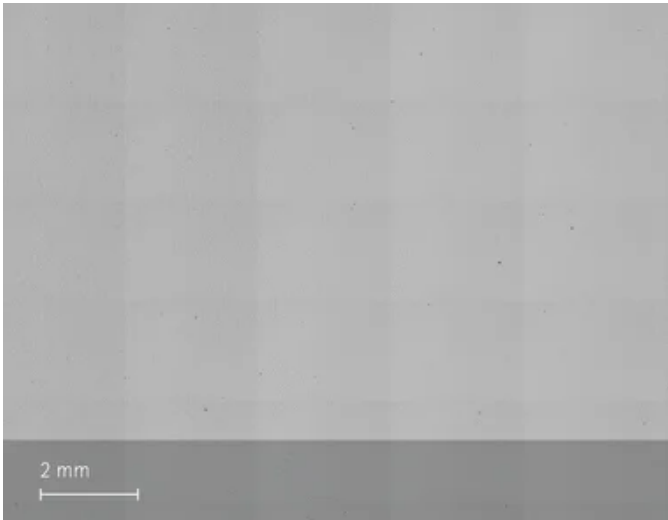
System Setup	EOS M 400-4
EOS Material set	PH1_040_080_Core M404
Software Requirements	EOSPRINT 2.11 or newer EOSYSTEM 2.14 or newer
Recoater Blade	Ceramic
Inert gas	Nitrogen
Sieve	63 µm

Additional Information	
Layer Thickness	40 µm, 80 µm, 40/80 µm
Volume Rate	4.1-7.9 mm³/s
Typical Dimensional Change after HT [%]	+0.1 %

Chemical and Physical Properties of Parts



Micrograph of polished surface. As manufactured 40µm



Micrograph of polished surface. As manufactured 80µm

Microstructure of the Produced Parts

Defects	Thickness	Result	Number of Samples
Average Defect Percentage	40 µm	<0.1 %	-
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Mechanical Properties

Mechanical Properties Heat Treated

EN ISO 6892-1 Room Temperature 40 µm	Yield Strength [MPa]	Tensile Strength [MPa]	Elongation at Break A [%]	Reduction of Area Z [%]	Modulus of elasticity [GPa]	Number of Samples
Vertical	1290	1430	8	-	-	-
Horizontal	1330	1450	8	-	-	-

H900 heat treatment

Mechanical Properties Heat Treated

EN ISO 6892-1 Room Temperature 80 µm	Yield Strength [MPa]	Tensile Strength [MPa]	Elongation at Break A [%]	Reduction of Area Z [%]	Modulus of elasticity [GPa]	Number of Samples
Vertical	1305	1440	7	-	-	-
Horizontal	1320	1460	7	-	-	-

H900 heat treatment

Impact Toughness

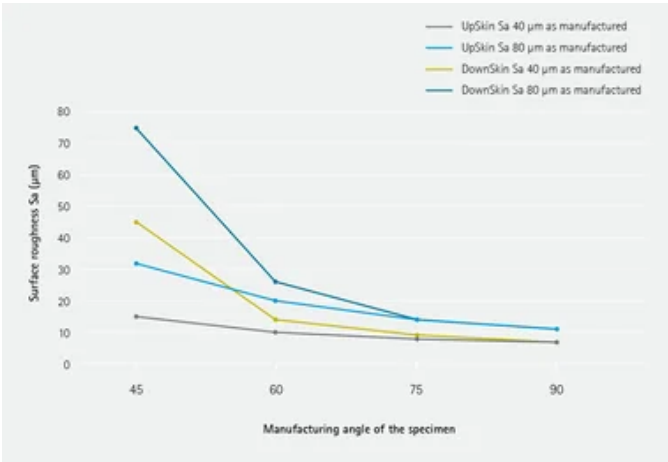
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Typical Impact Toughness [J]	8
Layer Thickness	40 µm

H900 heat treatment

EN ISO 148-1, Charpy-V Heat Treated	
Typical Impact Toughness [J]	6
Layer Thickness	80 µm

H900 heat treatment

Surface Roughness



Coefficient of Thermal Expansion

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10.6*10 ⁻⁶ /K	25 – 100 °C
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Status as of 18.08.2026. Subject to technical modifications. EOS is certified according to ISO 9001.

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